



Unit 4 · Outcome 1 · Climate Change

Responsible decision-making on climate change

KK13

VCE Environmental Science · Units 3 & 4 · Exam study summary

The whole chapter has been building to one hard question: now that we know what's coming, what should we actually do? This dot point is about the interconnections and tensions between the four forces that shape every real climate decision — stakeholder values, regulatory frameworks, scientific data and new technologies.

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



01 The four factors VCAA wants you to name

Every climate decision is pulled on by four forces. The study design names them exactly — learn these four words, because a question can simply ask you to apply them:

The beliefs and priorities of everyone affected — homes, jobs, Country, nature, future generations. Maps onto the value systems from U3·O1 (anthropocentric → ecocentric).

Treaties, laws, targets and planning rules that permit, require or forbid options — from the Paris Agreement down to a local planning overlay.

Observations, measurements and models — local tide-gauge records, IPCC projections, and the confidence ratings attached to them (very high → very low, from KK09).

Emerging options that change the menu — offshore wind, grid batteries, living-shoreline engineering, blue-carbon restoration, and less-proven ideas like direct air capture.

Factor	At Pellan Bay it looks like...
Stakeholder values	Residents fear losing homes; farmers fear saltwater on paddocks; Traditional Owners hold cultural responsibility for sea-Country and the saltmarsh; BirdLife wants the shorebird habitat protected; young people ask about their coastline in 2070.
Regulatory frameworks	Paris Agreement & Ramsar Convention (international); Climate Change Act 2022 (Cth) net-zero-2050 & EPBC Act 1999 protecting the Ramsar marsh (federal); Climate Action Act 2017 (Vic) net-zero-2045, the Marine and Coastal Act 2018 and climate-aware planning scheme (state/local).
Scientific data	The observatory's 30-year tide record; IPCC AR6 sea-level projections with confidence ratings; CSIRO/ BoM State of the Climate and the Victorian Climate Science Report 2024; council coastal-hazard inundation maps.
New technologies	A proposed Wirreanda offshore wind zone (aligned with Victoria's legislated 2 GW-by-2032 offshore-wind target), grid-scale batteries, nature-based "living shoreline" design, and saltmarsh blue-carbon restoration.



Figure · Factor compared at a glance.

INTERCONNECTION

02 Interconnections and tensions

This is the heart of the dot point — and the single most common place students lose marks. The four factors don't sit in separate boxes. They pull on each other. The exam word "interconnections and tensions" is really two ideas:

One factor strengthens, justifies or enables another. Example: scientific data on sea-level rise justifies the regulatory target, which in turn funds and mandates new offshore-wind technology.

One factor blocks, contradicts or competes with another. Example: a regulation protecting the Ramsar marsh (EPBC Act) blocks the seawall technology that would save the homes — a rule and a tech option in direct conflict.

Each line joins two of the four factors. Some pairs reinforce each other (an interconnection); others clash (a tension). Pick a line to read the specific interaction at Pellan Bay — and watch the same pair flip between help and conflict depending on the option chosen.

✓ **TIP**

The sentence that earns marks Don't just list four factors. Name a pair and state the relationship: "The scientific data (high-confidence local SLR) strengthens the case for the regulatory adaptation requirement, but it is in tension with residents' values, who weigh keeping their homes now above a projected risk decades away." That single sentence shows interconnection and tension across three factors.

03 The Pellan Bay decision arena



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 **P-E-L WRITING**

The Shire has three live proposals. Select each one to see how it scores on the four factors, what it trades off, and whether it tackles the cause (mitigation) or the consequence (adaptation) of climate change. There is no perfect option — that's the point.

04 Stakeholders and the values they hold

"Stakeholder values" is factor ①, and it connects straight back to the value systems from Unit 3 (anthropocentrism, biocentrism, ecocentrism, technocentrism). Different stakeholders weigh the same evidence completely differently — which is exactly why decisions create tension.

EVIDENCE-BASED

05 What turns a decision into a responsible one

VCAA uses the word "responsible" deliberately. A decision isn't responsible just because a choice was made. Use this checklist as the structure for any "evaluate / justify the decision" answer:

Grounded in the best available scientific data, and honest about its confidence level and uncertainty — not cherry-picked.

Applies the precautionary principle: where there's a risk of serious or irreversible harm, lack of full certainty is not a reason to delay protective action.

Considers intragenerational equity (who pays now — ratepayers? residents?) and intergenerational equity (the coastline future students inherit).

Sits within the regulatory frameworks — you can't lawfully drown a Ramsar marsh — and follows genuine stakeholder consultation.

Openly states the trade-offs and who bears them, rather than hiding the losers of the decision.

Can be revised as new data and technology arrive — a staged plan with review points beats a one-shot, locked-in bet.

✓ **TIP**

The mature conclusion Top responses rarely say "Option A is simply best." They argue for a defensible, staged package: e.g. back the wind farm (mitigation) AND begin managed retreat with living-shoreline restoration (adaptation), reviewed every 5 years as the observatory's data and IPCC confidence improve. That weighs all four factors, applies the precautionary and intergenerational-equity principles, and stays adaptive — which is exactly what "responsible" means.

06 Command-word decoder

◆ **COMMAND WORDS**

This dot point is almost always tested with high-order command words. Read what each is really asking before you write — tap a word: Each one demands a different shape of answer. "Identify" wants a word; "evaluate" wants a weighed judgement with reasons. Matching your structure to the command word is worth easy marks.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

07 P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 **P·E·L WRITING**

For the longer "analyse / evaluate" responses, build each paragraph as Point → Evidence → Link. Here's a worked P·E·L for a 4-mark "analyse the tension between two factors" question: Scientific data and stakeholder values are in tension over the Pellan Bay seawall. The observatory's tide record and IPCC AR6 projections (high confidence) justify acting now on inundation, yet many residents place greater value on keeping their homes and the cost burden than on a risk decades away — so the science points one way and present-day values point the other. A responsible decision applies the precautionary principle and intergenerational equity to act despite this tension, while using transparent consultation so affected residents are treated fairly.

✓ **TIP**

For "evaluate" questions Add a fourth move after Link: a Judgement — "On balance, the staged retreat is the more responsible option because... ". Evaluate marks are reserved for a clear, reasoned verdict, not a fence-sit.

09 Practice questions

★ **PRACTICE & SELF-MARK**

Five VCAA-style questions · 20 marks total. Attempt each before revealing the responses. Mark yourself honestly — your score feeds the dock in the corner.

UNIT 4 · OUTCOME 1 · KEY KNOWLEDGE 13

Responsible decision-making on climate change

The whole chapter has been building to one hard question: now that we know what's coming, what should we actually do? This dot point is about the interconnections and tensions between the four forces that shape every real climate decision — stakeholder values, regulatory frameworks, scientific data and new technologies.

Throughout this chapter you've worked at the Cape Wirreanda Climate Observatory on Victoria's south-west coast. Just along the shore sits the township of Pellan Bay: a few hundred homes, a caravan park, the only coast road in or out, low-lying dairy paddocks, and a Ramsar-listed saltmarsh that fills each spring with migratory shorebirds. The observatory's own tide gauge has measured the sea creeping up the foreshore for thirty years, and storm surges now reach further inland every few winters.

The Shire of Wirreanda has to decide what to do — and it cannot please everyone. Drag the scene below to look around, then push sea level up to see why a decision is being forced.

Digital Vector · VCE Environmental Science study summary · Climate Change · KK13 · digitalvector.com.au
Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.